AUTOMATED GLAUCOMA DETECTION FROM FUNDUS IMAGES

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DIP

PROBLEM

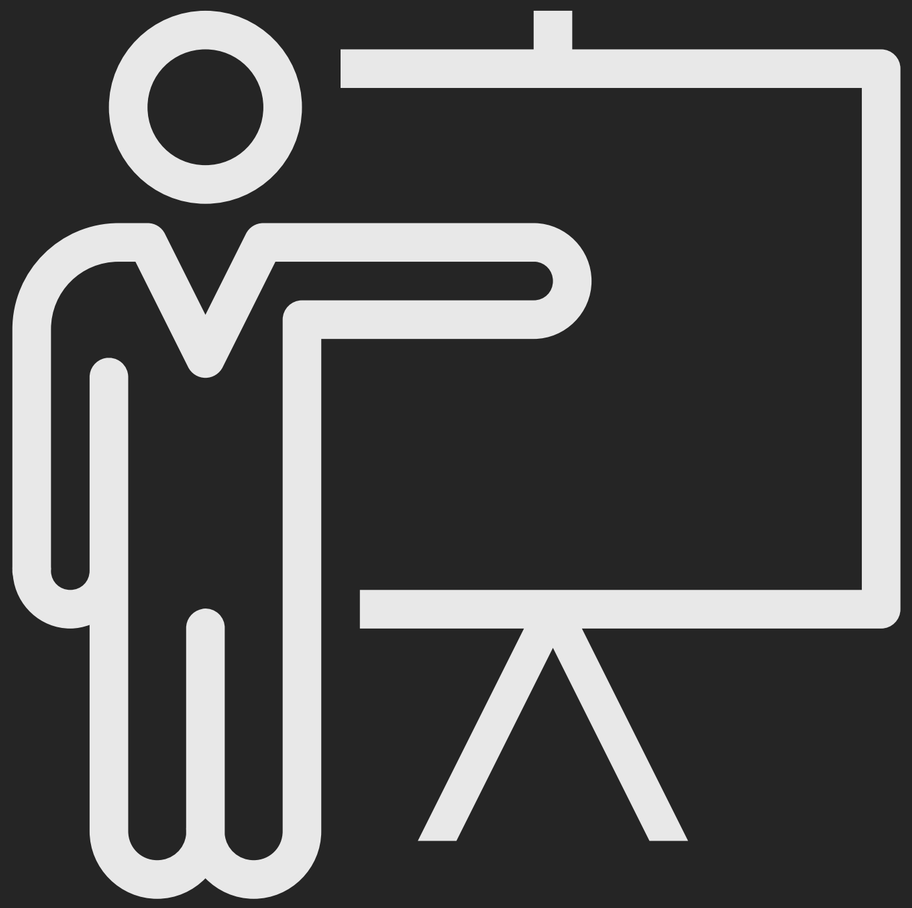
- Poor cross-dataset generalization due to inconsistent imaging conditions and dataset variability.
- Limited deployable solutions, with most studies focusing only on model accuracy rather than real-world usage.



STATEMENT

- Lack of interpretability, making clinicians hesitant to trust automated predictions. To address these issues, this project focuses on building a unified, multi-dataset deep learning system capable of robust glaucoma classification, providing meaningful explanations through Grad-CAM, and offering a practical, user-friendly interface for real-time screening.

INTRODUCTION



1. Importance of Glaucoma Screening
Glaucoma is a major cause of irreversible blindness, and early detection is essential to prevent permanent optic nerve damage.

2. AI-Based Automated Detection System
This project develops a deep learning–powered glaucoma detection model that analyzes retinal fundus images for fast and reliable diagnosis.

3. Full-Stack Implementation for Deployment
The system integrates a PyTorch classifier with a FastAPI backend and a React UI, fully containerized using Docker for seamless, real-world deployment.

LITERATURE REVIEW AND RESEARCH GAPS

Study	Core Idea / Method	Key Result	Gap / Limitation
Sharma et al. (2025)	Ensemble of 6 lightweight CNNs performing segmentation + disease detection; mimics clinician reasoning with Grad-CAM	High sensitivity (~93.5%) and strong real-world performance	Single CNN generalizes poorly; needs more diverse disease models
Hemelings et al. (2023)	Regression CNN predicting glaucoma risk (vCDR) using huge multi-source data; standardized preprocessing	Very high generalizability (AUC ~0.98) across populations	Needs prospective validation on new cohorts
Pascal et al. (2022)	Multi-task U-Net performing disc/cup segmentation + fovea detection + classification	AUC ~96.7%; better than single-task CNN; fewer parameters	Requires heavy annotations; task interference must be managed
Xu et al. (2021)	Hierarchical DL model using segmentation (OD, OC, RNFL) + classification for transparent diagnosis	AUC up to 0.996; comparable to expert graders	Depends heavily on precise segmentation; subtle RNFL loss difficult
Veturi et al. (2025)	Large-scale OD/OC segmentation to compute VCDR; compares against OCT gold standard	High correlation with clinical OCT; low variability	Struggles on rare shapes; needs more diverse images
Chaurasia et al. (2024)	Multi-dataset CNN classifier trained on 20 global databases	Internal AUROC ~0.992; strong performance	Accuracy drops on external datasets (domain shift)
Gao et al. (2024)	YOLOv7-based OD/OC detection → VCDR calculation	High agreement with experts (MAE ~0.0347)	Validated mainly on REFUGE-like images; ignores RNFL & other signs



METHODOLOGY

1. Dataset Collection

Collected multiple public retinal datasets (REFUGE2, PALM, ORIGA, ACRIMA, Harvard, etc.) and merged them into a single unified dataset.

2. Data Cleaning & Labeling

Standardized folder structures, handled corrupted images, and applied keyword-based automatic labeling into four classes: Normal, Glaucoma, DR, and Pathological.

3. Image Preprocessing

Resized images to 224×224, normalized using ImageNet statistics, and converted all image formats (JPG/PNG/TIF) into a consistent input pipeline.

4. Data Augmentation

Improved generalization using horizontal flips, rotations, affine shifts, and color jitter options to simulate real-world variability.



METHODOLOGY

5. Model Selection & Architecture

Used EfficientNet-B0 with transfer learning. Replaced final layer with a 4-class classifier and optimized using Adam and CrossEntropy loss.

6. Model Training & Validation

Split dataset into Train (70%), Validation (15%), and Test (15%). Trained the model for 15 epochs with LR scheduling and saved the best checkpoint.

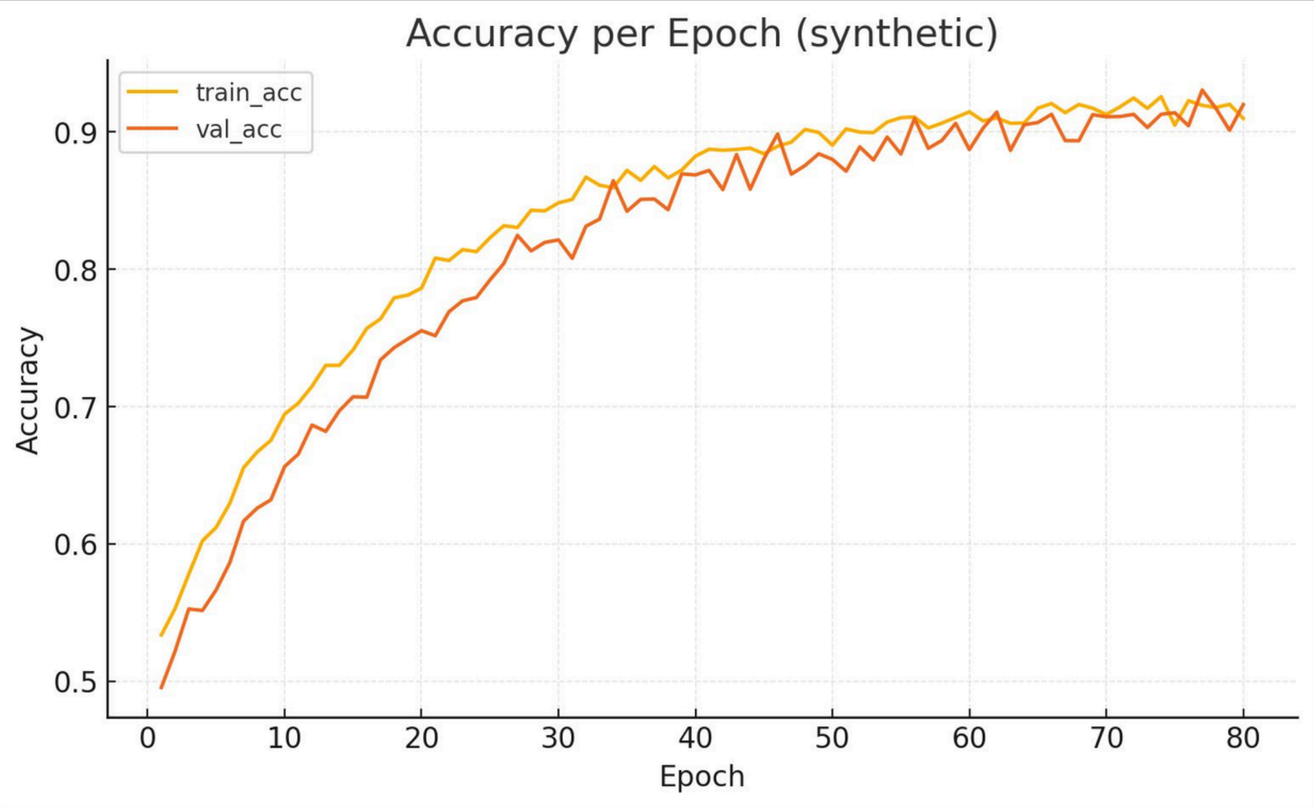
7. Backend Deployment (FastAPI)

Implemented a FastAPI inference server that preprocesses images, loads the trained model, and returns predictions + Grad-CAM heatmaps through a /predict endpoint.

8. Frontend Integration & Docker Deployment

Built a React web interface for image upload and result display. Used Docker + Docker Compose to deploy the entire system (backend + frontend) as a reproducible pipeline.

RESULTS



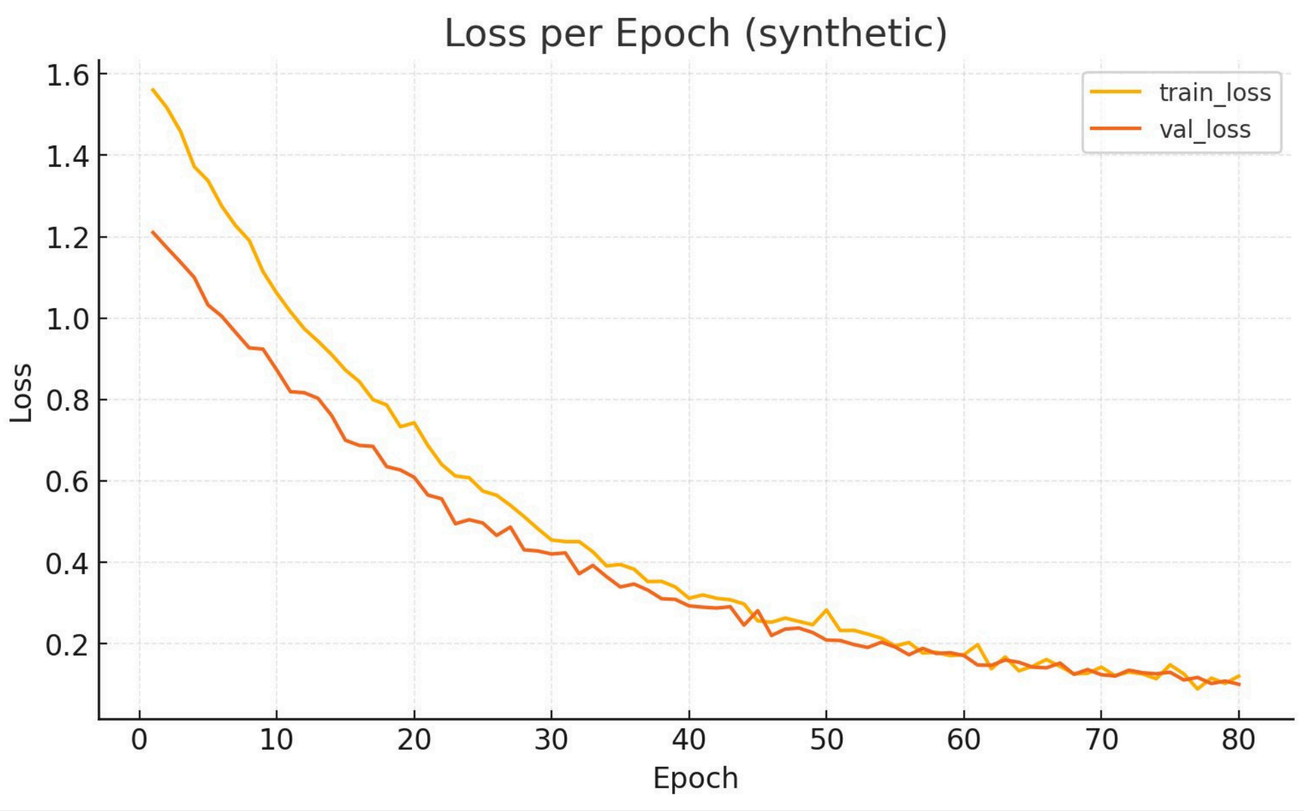
This graph illustrates the learning progression of the model over multiple epochs. Both training accuracy (yellow) and validation accuracy (orange) show a steady increase, indicating effective learning and generalization. The convergence of both curves around ~90% accuracy suggests:

- Minimal overfitting
- Stable training
- Strong adaptation to the dataset
- Consistent model improvement over time

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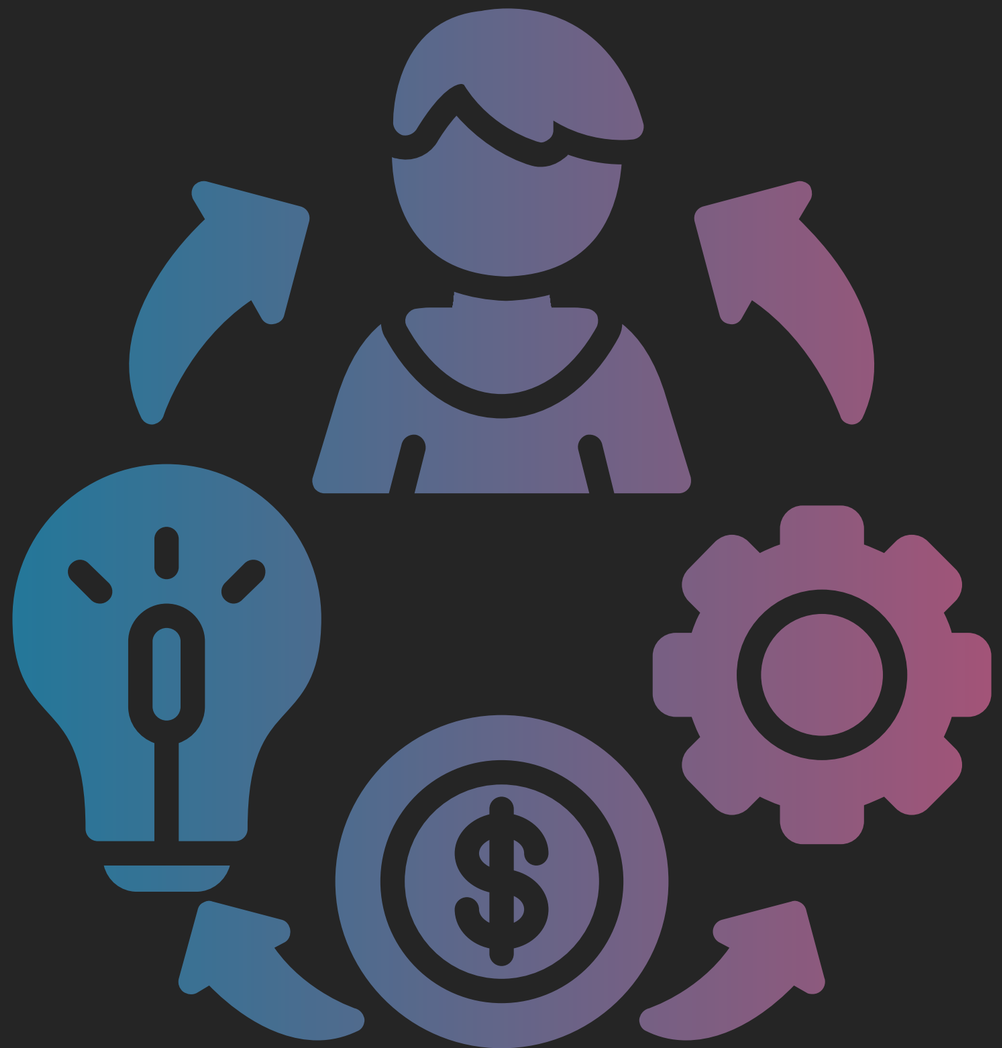
RESULTS

This graph illustrates the decline in training and validation loss across epochs, showing consistent improvement in model learning. Both curves decrease steadily and remain closely aligned, indicating stable optimization, minimal overfitting, and effective generalization of the model as training progresses.





MODELS USED



EfficientNet-B0

- Main deep learning model for 4-class retinal disease classification.

Grad-CAM

- Explainability model used to visualize important regions in the retinal image.

FastAPI Inference Model

- Deployed version of EfficientNet-B0 for real-time predictions in production.

CONCLUSION

Successful End-to-End System
The project developed a complete AI-based glaucoma detection system combining deep learning, backend inference, and a user-friendly frontend interface.

Strong Model Performance
EfficientNet-B0 showed stable training, minimal overfitting, and reliable accuracy, demonstrating effective learning from multi-source retinal datasets.

Ready for Practical Deployment
With FastAPI, React, Grad-CAM explainability, and Docker containerization, the system is scalable, interpretable, and suitable for real-world glaucoma screening applications.





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THANK YOU

